UDL Beam

# Example 1 - rivt Doc

Attn: User Example

proj. 0001

## Table of Contents

## 0.1 | Summary and Loads

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam using E-B theory [0.1.1] . It also serves as an annotated example of a single rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual .

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command at the end of each rivt file (Doc-API rv.D). Published files are found in the _published folder.

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### 0.1 - 2 | Load Combinations

Dead and live loads effects are taken from ASCE 7-05 [0.1.2]

**Table 1**: Load Effects (stored: t001-1.csv)

| Equation No. | Load Combination |
|---|---|
| 16-1 | 1.4(D+F) |
| 16-2 | 1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R) |
| 16-3 | 1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W) |

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### 0.1 - 3 | Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

**Table 2**: Define Unit Loads

| variable | value | [value] | description |
|---|---|---|---|
| D_1 | 3.80 p_sf | 0.18 kPA | joists DL |
| D_2 | 2.10 p_sf | 0.10 kPA | plywood DL |
| D_3 | 10.00 p_sf | 0.48 kPA | partitions DL |
| D_4 | 3.00 k_ft | 43.78 kN_m | fixed machinery DL |
| L_1 | 40.00 p_sf | 1.92 kPA | ASCE7-O5 LL |
| b_1 | 10.00 inch | 254.00 mm | beam width |

| variable | value | [value] | description |
|---|---|---|---|
| h_1 | 18.00 inch | 457.20 mm | beam depth |
| E_1 | 29000.00 k_si | 199947.96 MPA | modulus of elasticity |
| Fb_1 | 20000.00 p_si | 137.90 MPA | allowable stress |

The VALTABLE command reads variable values from a file in the rvsrc folder. The description is the table title, followed by the max column width.

**Table 3**: Beam Geometry (rvsrc/beam1.csv)

| variable | value | [value] | description |
|---|---|---|---|
| spc_1 | 2.00 ft | 0.61 m | beam spacing |
| spn_1 | 16.00 ft | 4.88 m | beam span |

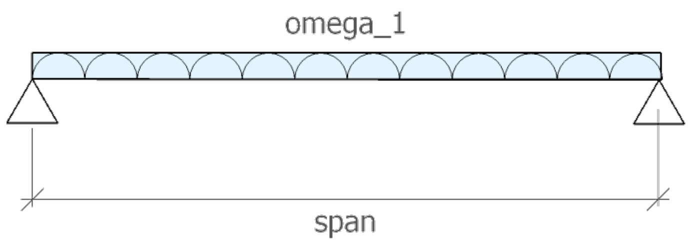


**Fig. 1** - Beam Diagram

**Uniform Distributed Loads**

**Eq. 1:** Dead load [ASCE7-05 2.3.2]

$$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$$

$dl_1$ = 3.64 k_ft     [$dl_1$] = 53.09 kN_m   | Dead load [ASCE7-05 2.3.2]

| $spc_1$ | $D_2$ | $D_4$ | $D_3$ | $D_1$ |
|---|---|---|---|---|
| 2.00 ft | 2.10 p_sf | 3.00 k_ft | 10.00 p_sf | 3.80 p_sf |
| beam spacing | plywood DL | fixed machinery DL | partitions DL | joists DL |

**Eq. 2:** Live load [ASCE7-05 2.3.2]

$$ll_1 = 1.6 \cdot L_1 \cdot spc_1$$

$ll_1$ = 0.13 k_ft [$ll_1$] = 1.87 kN_m | Live load [ASCE7-05 2.3.2]

| $spc_1$ | $L_1$ |
| --- | --- |
| 2.00 ft | 40.00 p_sf |
| beam spacing | ASCE7-05 LL |

**Eq. 3:** Total load [ASCE7-05 2.3.2]

$$\omega_1 = dl_1 + ll_1$$

$\omega_1$ = 3.77 k_ft [$\omega_1$] = 54.96 kN_m | Total load [ASCE7-05 2.3.2]

| $dl_1$ | $ll_1$ |
| --- | --- |
| 3.64 k_ft | 128.00 ft·p_sf |
| Dead load [ASCE7-05 2.3.2] | Live load [ASCE7-05 2.3.2] |

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## 0.1 - 4 | Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

**Table 4**: Beam functions (rvsrc/scripts/sectprop.py)

| Function | Docstring |
| --- | --- |
| rectsect(b, d) | section modulus of rectangle |
| rectinertia(b, d) | moment of inertia of rectangle |
| midspan_delta(ln, w, e, i) | mid-span deflection of simply supported beam with UDL |

**Eq. 4:** rectangle - S (sectprop.py)

$$section_1 = rectsect(b_1, h_1)$$

$section_1$ = 540.00 in3 [$section_1$] = 8849.01 cm3 | rectangle - S (sectprop.py)

| $b_1$ | $h_1$ |
|---|---|
| 10.00 inch | 18.00 inch |
| beam width | beam depth |

**Eq. 5:** rectangle - I (sectprop.py)

```
inertia₁ = rectinertia(b₁, h₁)

inertia₁ = 4860.0 in4     [inertia₁] = 202288.5 cm4   | rectangle - I (sectprop.py)
```

| $b_1$ | $h_1$ |
|---|---|
| 10.0 inch | 18.0 inch |
| beam width | beam depth |

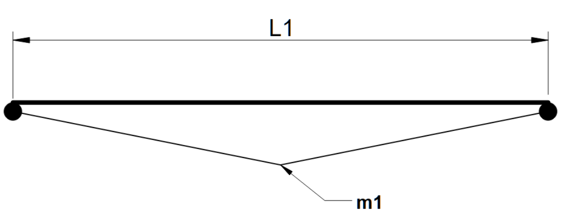


**Fig. 2 -** Moment diagram

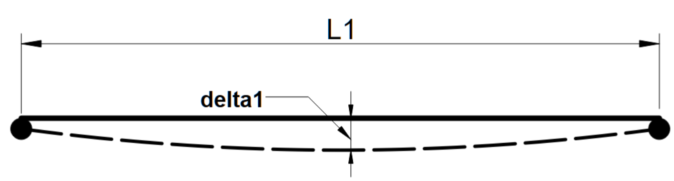


**Fig. 3 -** Deflection diagram

**Maximum bending stress formula**

**Eq.6:**

$$\sigma_1 = \frac{M_1}{S_1}$$

**Eq. 7:** Mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot spn_1^2}{8}$$

$m_1$ = 120.52 ft-kip [$m_1$] = 163.40 mkN | Mid-span UDL moment

| $spn_1$ | $\omega_1$ |
|---|---|
| 16.00 ft | 3.77 k_ft |
| beam span - | Total load [ASCE7-05 2.3.2] |

**Eq. 8:** Bending stress

$$fb_1 = \frac{m_1}{section_1}$$

$fb_1$ = 2678.2 p_si [$fb_1$] = 18.5 MPA | Bending stress

| $section_1$ | $m_1$ |
|---|---|
| 540.0 inch3 | 120.5 ft2·k_ft |
| rectangle - S (sectprop.py) | Mid-span UDL moment - |

**Eq.9:** Stress ratio

▮ $fb_1 < Fb_1$

| [1] $fb_1$ | [2] $Fb_1$ | ratio [1]/[2] | check | reference |
|---|---|---|---|---|
| 2.68 k_si | 20.00 k_si | 0.13 | OK | Stress ratio |

**Eq. 10:** mid-span deflection (sectprop.py)

$\delta_1$ = midspan_δ($spn_1$, $\omega_1$, $E_1$, $inertia_1$)

$\delta_1$ = 0.04 inch [$\delta_1$] = 1.00 mm | mid-span deflection (sectprop.py)

| $E_1$ | $inertia_1$ | $spn_1$ | $\omega_1$ |
|---|---|---|---|
| 29000.00 k_si | 4860.00 inch4 | 16.00 ft | 3.77 k_ft |
| modulus of elasticity | rectangle - I (sectprop.py) | beam span - | Total load [ASCE7-05 2.3.2] |

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**[0.1.1]**

"Euler–Bernoulli beam theory", Wikipedia, Wikimedia Foundation. [Online].https://en.wikipedia.org/wiki/Euler_Bernoulli_beam_theory.[Accessed: Jun. 15, 2026].

**[0.1.2]**

ASCE/SEI 7-05, Minimum Design Loads for Buildings and Other Structures,American Society of Civil Engineers, 2005.